

Dwell-Conditioned Cycle-to-Cycle Switching-Time Feedback for Mode-Dependent Switched-Affine Systems

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Abstract

Static cycle-to-cycle feedback regulates a reference periodic trajectory by shifting the interior switching instants of a fixed-period mode sequence. Its raw timing action can nevertheless produce inadmissibly short or negative dwell durations. We derive the one-cycle model for mode-dependent switched-affine dynamics and condition the complete raw action by its largest dwell-feasible scalar factor. The conditioner preserves the action direction, mode order, and cycle period and requires an $\mathcal{O}(N)$ scan without an online solver. Uniform scaling places the conditioned linearized matrices on a segment between the nominal cycle matrix and the raw closed-loop matrix. An offline semidefinite program seeks a common quadratic Lyapunov function at these two endpoints. We prove that strict endpoint decrease implies uniform exponential stability for every sequence of conditioning factors in $[0, 1]$. A three-cell DC–DC converter benchmark admits this certificate. Exact switched-affine simulations compare the controlled and open-loop three-dimensional trajectories and show eventual open-loop convergence to the same reference trajectory. Conditioned feedback reaches a normalized error below **0.01** at cycle 6, compared with persistent open-loop crossing at cycle 22009, while enforcing the assumed 3 μ s dwell bound. The common Lyapunov result concerns the linearized family; exact nonlinear stability remains local, and the large-error response is simulation evidence.

Keywords: switched-affine systems, periodic trajectories, switching-time control, dwell-time conditioning, common quadratic Lyapunov function, semidefinite programming

1 Introduction

Power converters control voltage and current by moving through a sequence of switch configurations. In practice, the switching pattern must respect electrical limits, switching frequency, a minimum dwell duration for each configuration, and a short computation time [Karamanakos et al. \(2020\)](#). Keeping the individual configurations explicit gives a hybrid-affine or switched-affine model [Albea-Sanchez et al. \(2021\)](#). Repeating the pattern produces a periodic trajectory rather than a

constant state, and controlling such trajectories is an established problem for switched-affine systems [Egidio et al. \(2020\)](#); [Xu and Lazar \(2024\)](#). We consider regulation at a selected cycle anchor. At each cycle start, the controller steers the state toward that point on the reference periodic trajectory.

Predictive control tackles the problem by solving a constrained optimization problem online. Recent work reduces the online burden through Laguerre parameterization, dedicated fixed-frequency solvers, and candidate pruning [Saeed et al. \(2022\)](#); [Yang et al. \(2022\)](#); [Augustine and Patil \(2023\)](#).

Controllers also differ in what they may change. State-dependent switching laws choose the active subsystem [Egidio et al. \(2022\)](#), where switching-time control keeps an ordered sequence and moves its transition instants. We take this same approach, changing the transition instants and keeping the rest of the parameters the same. At each cycle start, the error is computed and the offsets applied during that cycle, with no online solver.

Patino et al. studied mode-dependent cyclic systems and how each switching instant affects the trajectory. They applied an online predictive controller to adjust the switching instants and track an optimal periodic trajectory [Patino et al. \(2010\)](#). Marcolino et al. considered a fixed-period cycle controlled by changing its interior switching instants. They used a discrete LQR and dwell-constrained predictive control to apply corrections based on the one-cycle model [Marcolino et al. \(2021\)](#). Their model has a common state matrix with switched actuator levels. Each interior switching instant separates two dwell durations, corresponding to the modes active before and after it. Delaying the instant lengthens the preceding dwell and shortens the following one, so the dwell limits are coupled. The predictive controller enforces these limits through online optimization, and its feasible set is characterized through polyhedral projection [Marcolino et al. \(2021\)](#). The standalone LQR comparison instead clips the offsets one at a time, which changes the requested direction and may shift the final scheduled instant [Marcolino et al. \(2021\)](#). Projecting a raw action directly onto the feasible set would require a quadratic program and would generally change its direction. Direction-preserving control allocation scales the complete command, but it also uses online optimization and does not address fixed-period dwell constraints [Nakamura-Zimmerer and Miller \(2026\)](#). The remaining problem is to find the largest feasible scale directly, preserving the timing-action direction, mode order, and cycle period without an online solver.

We adapt direction-preserving scaling to fixed-period switching schedules. Each dwell constraint gives an upper bound on the scale, and the smallest bound is the largest feasible factor in $[0, 1]$. Checking all N bounds takes $O(N)$ time and requires no online solver. Scaling the complete raw offset vector preserves its direction, the mode order, and the cycle period. For the linearized raw

closed loop, an offline polyhedral iteration finds the maximal invariant subset of the Raw-Action Admissible Region. Outside this set, the conditioner guarantees current-cycle dwell feasibility but not convergence; the nonlinear guarantee remains local. [Section 2](#) derives the cycle model, [section 3](#) develops the conditioner and invariant region, and [section 5](#) presents the converter study.

2 Problem formulation and one-cycle model

The control objective is to approach a prescribed periodic trajectory without changing its mode sequence or cycle period. We first express this objective at cycle starts, then derive how shifts of the interior switching instants change the next cycle-start state.

2.1 Plant, reference trajectory, and timing coordinates

Consider the switched-affine plant

$$\dot{\mathbf{x}}(t) = A_{\sigma(t)}\mathbf{x}(t) + \mathbf{b}_{\sigma(t)}, \quad \mathbf{x}(t) \in \mathbb{R}^n. \quad (1)$$

Every cycle visits the fixed sequence $(q_1, \dots, q_N)$. Its nominal cycle-relative boundaries and dwell durations are

$$\begin{aligned} 0 = \bar{t}_0 < \bar{t}_1 < \dots < \bar{t}_N = T, \\ \bar{d}_i = \bar{t}_i - \bar{t}_{i-1}. \end{aligned} \quad (2)$$

Repeating this schedule generates the reference periodic trajectory $\bar{\mathbf{x}}(t)$. Its cycle anchor is $\mathbf{x}^* = \bar{\mathbf{x}}(0) = \bar{\mathbf{x}}(T)$. With $\mathbf{x}_k = \mathbf{x}(kT)$, the cycle-start error is $\mathbf{e}_k = \mathbf{x}_k - \mathbf{x}^*$. The target is this phase-aligned trajectory, not a constant average operating point.

The controller moves only the $N - 1$ interior instants:

$$\begin{aligned} t_{k,i} &= \bar{t}_i + \delta\tau_{k,i}, \quad i = 1, \dots, N - 1, \\ \delta\tau_{k,0} &= \delta\tau_{k,N} = 0. \end{aligned} \quad (3)$$

During interval i of cycle k , the plant uses mode q_i between the absolute times $kT + t_{k,i-1}$ and

$kT + t_{k,i}$. The applied dwell vector satisfies

$$\begin{aligned} \mathbf{d}_k &= \bar{\mathbf{d}} + \delta \mathbf{d}_k, & \delta \mathbf{d}_k &= D \delta \boldsymbol{\tau}_k, \\ D_{ij} &= \begin{cases} 1, & i = j, \\ -1, & i = j + 1, \\ 0, & \text{otherwise,} \end{cases} \end{aligned} \quad (4)$$

where $D \in \mathbb{R}^{N \times (N-1)}$. Thus delaying one instant lengthens the preceding interval and shortens the following one by the same amount. Since $\mathbf{1}^\top D = 0$, every offset vector preserves T . The remaining requirement is $\mathbf{d}_k \geq d_{\min} \mathbf{1}$ for a specified $d_{\min} > 0$.

2.2 Exact one-cycle map

Introduce the augmented state, interval generator, and nominal transition matrix

$$\begin{aligned} \chi &= \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix}, & F_i &= \begin{bmatrix} A_{q_i} & \mathbf{b}_{q_i} \\ \mathbf{0}^\top & 0 \end{bmatrix}, \\ E_i &= e^{F_i \bar{d}_i}. \end{aligned} \quad (5)$$

On interval i , $\dot{\chi} = F_i \chi$, so the exact cycle propagation is

$$\chi_{k+1} = e^{F_N d_{k,N}} \dots e^{F_1 d_{k,1}} \chi_k. \quad (6)$$

The rightmost factor acts first; matrices belonging to different modes cannot generally be interchanged. Define

$$\Pi = [I_n \ \mathbf{0}], \quad \iota = \begin{bmatrix} I_n \\ \mathbf{0}^\top \end{bmatrix}. \quad (7)$$

The physical one-cycle map is

$$\begin{aligned} \mathcal{P}(\mathbf{x}, \delta \boldsymbol{\tau}) &= \Pi e^{F_N [\bar{d}_N + (D \delta \boldsymbol{\tau})_N]} \dots \\ &\quad e^{F_1 [\bar{d}_1 + (D \delta \boldsymbol{\tau})_1]} \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix}. \end{aligned} \quad (8)$$

The reference schedule and anchor satisfy $\mathcal{P}(\mathbf{x}^*, \mathbf{0}) = \mathbf{x}^*$. They are given design data, not optimization variables of the timing controller.

2.3 Linearization in the independent switching instants

For a dwell change δd_i , the single-interval expansion is

$$e^{F_i (\bar{d}_i + \delta d_i)} = E_i + F_i E_i \delta d_i + \mathcal{O}(\delta d_i^2). \quad (9)$$

Expanding the ordered product, while retaining only terms containing one dwell change, gives

$$\begin{aligned} &e^{F_N (\bar{d}_N + \delta d_N)} \dots e^{F_1 (\bar{d}_1 + \delta d_1)} \\ &= \tilde{\Phi} + \sum_{i=1}^N J_i \delta d_i + \mathcal{O}(\|\delta \mathbf{d}\|^2), \\ &\tilde{\Phi} = E_N \dots E_1, \\ &J_i = E_N \dots E_{i+1} F_i E_i \dots E_1. \end{aligned} \quad (10)$$

An empty product is the identity. In particular, $J_1 = E_N \dots E_2 F_1 E_1$ and $J_N = F_N E_N \dots E_1$. Write $\chi_k = \chi^* + \iota \mathbf{e}_k$, with $\chi^* = [(\mathbf{x}^*)^\top \ 1]^\top$. Since $\tilde{\Phi} \chi^* = \chi^*$, subtracting the nominal propagation cancels the zeroth-order term. The products $\delta d_i \mathbf{e}_k$ and $\delta d_i \delta d_j$ are second order, leaving

$$\begin{aligned} \chi_{k+1} - \chi^* &= \tilde{\Phi} \iota \mathbf{e}_k + \sum_{i=1}^N J_i \chi^* \delta d_{k,i} \\ &\quad + \mathcal{O}((\|\mathbf{e}_k\| + \|\delta \mathbf{d}_k\|)^2). \end{aligned} \quad (11)$$

Projecting onto the physical state and using $\delta \mathbf{d}_k = D \delta \boldsymbol{\tau}_k$ yields

$$\begin{aligned} \mathbf{e}_{k+1} &= \Phi \mathbf{e}_k + \Gamma_\tau \delta \boldsymbol{\tau}_k + \mathcal{O}((\|\mathbf{e}_k\| + \|\delta \boldsymbol{\tau}_k\|)^2), \\ \Phi &= \Pi \tilde{\Phi} \iota = e^{A_{q_N} \bar{d}_N} \dots e^{A_{q_1} \bar{d}_1}, \\ G_d &= [\Pi J_1 \chi^* \dots \Pi J_N \chi^*], \\ \Gamma_\tau &= G_d D. \end{aligned} \quad (12)$$

The remainder is understood in fixed state and timing units. The augmented product has the block form

$$\tilde{\Phi} = \begin{bmatrix} \Phi & \mathbf{h} \\ \mathbf{0}^\top & 1 \end{bmatrix}. \quad (13)$$

Its unit eigenvalue belongs to the auxiliary constant coordinate. Stability is tested on the physical error dynamics, not on this augmented matrix.

Each column of D subtracts two adjacent dwell sensitivities. Factoring them gives

$$\begin{aligned} \Gamma_{\tau,i} &= e^{A_{q_N} \bar{d}_N} \dots e^{A_{q_{i+1}} \bar{d}_{i+1}} \\ &\quad \times [(A_{q_i} - A_{q_{i+1}}) \bar{\mathbf{x}}(\bar{t}_i) + \mathbf{b}_{q_i} - \mathbf{b}_{q_{i+1}}] \end{aligned} \quad (14)$$

The bracket is the change in the affine vector field caused by trading time between adjacent modes. The remaining transitions carry this effect to the cycle end. This recovers the switching-sensitivity structure in [Patino et al. \(2010\)](#); switching-time

sensitivities also appear in the hybrid limit-cycle analysis of [Flieller et al. \(2006\)](#).

For the special case $A_{q_i} = A_c$ and $\mathbf{b}_{q_i} = B_c \mathbf{u}_{i-1}$,

$$\Phi = e^{A_c T}, \quad \Gamma_{\tau,i} = e^{A_c(T-\bar{t}_i)} B_c (\mathbf{u}_{i-1} - \mathbf{u}_i), \quad (15)$$

which is the fixed-period switched-actuator model of [Marcolino et al. \(2021\)](#), with the corresponding input-jump sign convention. Mode-dependent state matrices require the full expression in (14).

2.4 Raw proportional timing feedback

A fixed gain $K \in \mathbb{R}^{(N-1) \times n}$ requests

$$\delta \tau_k^{\text{raw}} = -K \mathbf{e}_k, \quad A_{\text{cl}} = \Phi - \Gamma_{\tau} K. \quad (16)$$

The gain can be designed from (12), for example by discrete LQR. Here the gain is fixed before conditioning and certification. Although a Schur A_{cl} gives a locally stabilizing raw law, it does not make every requested schedule admissible. The next section enforces the dwell constraints; [section 4](#) examines the resulting conditioned dynamics.

3 Solver-free dwell-time conditioning

Minimum dwell durations couple adjacent offsets through D . Independent clipping need not preserve the direction of the feedback action. Instead, we apply one scalar factor to the complete raw vector.

Assume that every nominal interval has strict margin,

$$\mathbf{m} = \bar{\mathbf{d}} - d_{\min} \mathbf{1} > \mathbf{0}. \quad (17)$$

The Raw-Action Admissible Region is

$$\mathcal{C} = \{\mathbf{e} : DK\mathbf{e} \leq \mathbf{m}\}. \quad (18)$$

Inside $\mathcal{C}$, the raw action already satisfies all dwell constraints. Outside it, at least one requested interval is too short. With $\mathbf{q}_k = D\delta \tau_k^{\text{raw}}$, define

$$\beta_k = \min \left\{ 1, \min_{i: q_{k,i} < 0} \frac{m_i}{-q_{k,i}} \right\}, \quad (19)$$

$$\delta \tau_k^{\text{app}} = \beta_k \delta \tau_k^{\text{raw}},$$

where a minimum over an empty set is $+\infty$. Only intervals that the raw action would shorten impose upper bounds on β_k .

Proposition 1 (Feasibility and maximality). *Under (17), (19) returns the largest $\beta \in [0, 1]$ for which every applied dwell satisfies $\bar{\mathbf{d}} + \beta D \delta \tau_k^{\text{raw}} \geq d_{\min} \mathbf{1}$. For a finite raw action, $\beta > 0$. The conditioner preserves the direction of a nonzero raw action, mode order, and cycle period. Computing the factor takes $O(N)$ time without an online solver.*

Proof For $q_{k,i} \geq 0$, $m_i + \beta q_{k,i} > 0$ for all $\beta \in [0, 1]$. For $q_{k,i} < 0$, feasibility is equivalent to $\beta \leq m_i / (-q_{k,i})$. Intersecting these bounds with $[0, 1]$ gives (19), proving feasibility and maximality. Every finite bound is positive. Uniform positive scaling preserves direction, and $\mathbf{1}^T D = 0$ preserves T . Since $d_{\min} > 0$, the boundaries remain ordered. Adjacent differences compute $\mathbf{q}_k$, and one scan evaluates the N bounds. $\square$

At each cycle start, the controller therefore measures $\mathbf{x}_k$, computes the raw action, scans the dwell changes, and applies the scaled offsets. If $\beta_k < 1$, at least one dwell reaches its minimum. No set-membership test or optimization is required online. The polyhedron $\mathcal{C}$ explains where conditioning is inactive; its maximal invariant subset is treated separately in [section A](#).

4 Lyapunov stability and convex optimization

Feasibility alone does not imply convergence. This section gives a sufficient stability condition for the complete conditioned linearized family, including cycles in which $\beta_k < 1$.

4.1 Conditioned matrix family

Let S_x be a fixed positive diagonal state-scale matrix and set $\mathbf{z}_k = S_x^{-1} \mathbf{e}_k$. Define the normalized endpoints

$$A_0 = S_x^{-1} \Phi S_x, \quad A_1 = S_x^{-1} A_{\text{cl}} S_x. \quad (20)$$

Substituting (19) into the first-order model gives

$$\mathbf{z}_{k+1} = A(\beta_k) \mathbf{z}_k, \quad (21)$$

$$A(\beta) = (1 - \beta) A_0 + \beta A_1.$$

Equation (21) omits the nonlinear remainder in (12). Although the implemented conditioner returns $\beta > 0$ for finite errors, certifying the closed interval $[0, 1]$ includes its limiting zero-feedback endpoint. Similarity by S_x changes neither endpoint's eigenvalues.

Schur stability of A_0 and A_1 separately is insufficient to establish stability of every matrix in the segment or of their time-varying products. We seek one quadratic function that decreases for both endpoints.

4.2 Common quadratic Lyapunov certificate

Theorem 1 (Uniform stability under conditioning). *Suppose there exist $P = P^\top \succ 0$ and $\eta > 0$ such that*

$$P - A_i^\top P A_i \succeq \eta I_n, \quad i \in \{0, 1\}. \quad (22)$$

Then $V(\mathbf{z}) = \mathbf{z}^\top P \mathbf{z}$ decreases for every nonzero state of (21), uniformly over $\beta \in [0, 1]$. For any sequence $\{\beta_k\} \subset [0, 1]$,

$$\begin{aligned} \|\mathbf{z}_k\|_2 &\leq \sqrt{\frac{\lambda_{\max}(P)}{\lambda_{\min}(P)}} r^k \|\mathbf{z}_0\|_2, \\ r &= \sqrt{1 - \frac{\eta}{\lambda_{\max}(P)}} < 1. \end{aligned} \quad (23)$$

In particular, the result holds for the state-dependent factor in (19).

Proof For the same $P \succ 0$ and $\eta > 0$ at both endpoints, define

$$\mathcal{S}_\eta(\beta) = \begin{bmatrix} P - \eta I_n & A(\beta)^\top P \\ P A(\beta) & P \end{bmatrix}. \quad (24)$$

Since $P \succ 0$, the Schur complement of its lower-right block gives the equivalence

$$\mathcal{S}_\eta(\beta) \succeq 0 \iff P - A(\beta)^\top P A(\beta) \succeq \eta I_n.$$

The endpoint conditions therefore give $\mathcal{S}_\eta(0), \mathcal{S}_\eta(1) \succeq 0$. The affine dependence on β yields

$$\begin{aligned} \mathcal{S}_\eta(\beta) &= (1 - \beta)\mathcal{S}_\eta(0) \\ &\quad + \beta\mathcal{S}_\eta(1) \succeq 0 \end{aligned} \quad (25)$$

for every $\beta \in [0, 1]$, by convexity of the positive semidefinite cone. The Schur complement then gives, for $\mathbf{z}_{k+1} = A(\beta_k)\mathbf{z}_k$,

$$\begin{aligned} V(\mathbf{z}_{k+1}) - V(\mathbf{z}_k) &\leq -\eta \|\mathbf{z}_k\|_2^2 \\ &\leq -\frac{\eta}{\lambda_{\max}(P)} V(\mathbf{z}_k). \end{aligned} \quad (26)$$

The same bound holds independently at every cycle, without any assumption on how β_k is selected. Iteration and the inequalities $\lambda_{\min}(P)\|\mathbf{z}\|_2^2 \leq V(\mathbf{z}) \leq \lambda_{\max}(P)\|\mathbf{z}\|_2^2$ give (23). $\square$

The Schur-complement argument proves decrease on the entire matrix segment, not only at sampled values of β . It certifies the linearized conditioned model for all initial errors in that model, including those outside $\mathcal{C}$. It does not make the linear approximation valid at arbitrary physical errors.

4.3 Offline semidefinite program

For a fixed gain and fixed endpoint matrices, the certificate can be found by

$$\begin{aligned} &\underset{P=P^\top, \eta}{\text{maximize}} && \eta \\ &\text{subject to} && \text{tr}(P) = n, \\ & && P \succeq \varepsilon_P I_n, \\ & && P - A_0^\top P A_0 \succeq \eta I_n, \\ & && P - A_1^\top P A_1 \succeq \eta I_n, \end{aligned} \quad (27)$$

with $\varepsilon_P = 10^{-7}$. The trace constraint removes the arbitrary positive scale of P . For fixed A_0, A_1 , both these inequalities and their equivalent inverse-free block form $\mathcal{S}_\eta(0), \mathcal{S}_\eta(1) \succeq 0$ are affine in P, η . The Julia implementation uses the smaller inequalities in (27). The gain K is fixed, not jointly optimized. A stability certificate requires a positive verified decrease margin, not merely successful solver termination.

After solving, we symmetrize the returned P and recompute

$$\begin{aligned} Q_i &= P - A_i^\top P A_i, \quad i \in \{0, 1\}, \\ \mu &= \min\{\lambda_{\min}(Q_0), \lambda_{\min}(Q_1)\}. \end{aligned} \quad (28)$$

Positive definiteness, trace normalization, and $\mu > 0$ are checked directly. The verified margin μ can replace η in (23). Maximizing η under trace normalization does not directly minimize the resulting rate bound, which also depends on $\lambda_{\max}(P)$.

The certificate is sufficient, not necessary. In particular, including $\beta = 0$ requires A_0 to be Schur. An unstable open-loop endpoint cannot satisfy this full-interval test, even if a more restricted state-dependent controller might stabilize it. Failure to

obtain a positive common margin does not by itself establish instability.

4.4 What follows for the exact plant

The exact conditioned cycle map retains a state-dependent timing remainder. Since $\|\delta\tau^{\text{app}}\| \leq \|K\| \|e\|$, its normalized local form is

$$z_{k+1} = A(\beta_k)z_k + \mathcal{O}(\|z_k\|^2). \quad (29)$$

The common- P proof does not bound that remainder over a large-error region.

There is nevertheless a local exact result. Strict nominal dwell margin places a neighborhood of $e = \mathbf{0}$ inside $\mathcal{C}$. There the conditioner is inactive, and the exact closed-loop map is smooth with Jacobian A_{cl} . The endpoint inequalities imply that this Jacobian is Schur. The discrete-time linearization theorem therefore gives local exponential stability of the cycle anchor, in a sufficiently small invariant neighborhood. Smooth interval flows and switching offsets tending to zero extend convergence to the phase-aligned reference periodic trajectory between cycle starts. This argument does not claim nonlinear invariance of the entire polyhedron $\mathcal{C}$.

Open loop is a distinct case. With fixed nominal durations, (13) gives $\mathcal{P}(x, \mathbf{0}) = \Phi x + h$ exactly. Subtracting its fixed point yields

$$e_{k+1} = \Phi e_k \quad (30)$$

without a linearization remainder. Thus a Schur Φ proves convergence under the prescribed open-loop schedule throughout this ideal affine model. This fact does not extend to the nonlinear timing-feedback map.

5 Application to a three-cell converter

The application checks the cycle model, evaluates the common- P certificate, and compares exact controlled and open-loop trajectories. Since the benchmark is already open-loop stable, the practical question is how much feedback accelerates convergence while preserving dwell feasibility.

5.1 Benchmark and controller parameters

We use the three-cell multilevel DC–DC converter benchmark of Patino et al. [Patino et al. \(2010\)](#), with $x = [v_{C_1}, v_{C_2}, i_L]^T$. For binary switch commands $u_1, u_2, u_3 \in \{0, 1\}$,

$$A_u = \begin{bmatrix} 0 & 0 & (u_2 - u_1)/C_1 \\ 0 & 0 & (u_3 - u_2)/C_2 \\ (u_1 - u_2)/L & (u_2 - u_3)/L & -R/L \end{bmatrix},$$

$$b_u = \begin{bmatrix} 0 \\ 0 \\ Eu_3/L \end{bmatrix}. \quad (31)$$

Both matrices depend on the switch state, so the common-state-matrix reduction in (15) does not apply. The fixed nine-interval physical mode sequence is $(0, 1, 3, 7, 2, 0, 4, 7, 4)$, where the physical mode ID is $4u_1 + 2u_2 + u_3$.

The published cycle boundaries are $(0, 66, 88, 110, 132, 154, 220, 242, 264, 286)$ μs , with anchor $x^* = (9.9247, 19.2928, 0.9823)^T$. These rounded data do not close exactly. A one-time least-squares dwell reconciliation preserves the reported anchor, 286 μs period, and 22 μs nominal-design bound. It gives

$$\bar{t} = (0, 63.889, 87.557, 109.557, 132.538, 154.538, 218.210, 240.380, 263.828, 286) \mu\text{s}. \quad (32)$$

Computations use the full-precision schedule, with closure error 1.421e-14 in the infinity norm and maximum boundary correction 2.111 μs . This reconciliation prepares the benchmark; it is not a proposed reference-trajectory design method.

[Figure 1](#) shows the reference trajectory. Its cycle averages are 10.410 56 V, 19.422 28 V, and 0.956 13 A. With $v = \delta\tau/t_s$, the normalized timing-input matrix is $B_n = S_x^{-1}\Gamma_\tau t_s$. Discrete LQR on (A_0, B_n) yields $v^{\text{raw}} = -K_n z$, or $K = t_s K_n S_x^{-1}$ in physical coordinates. [Table 1](#) lists the weights. Only the aggressive law is conditioned; the conservative law is an unconditioned comparison that remains dwell-feasible for this experiment.

5.2 Model checks and numerical Lyapunov certificate

Central finite differences of the exact map check both Jacobians. State steps are 10^{-4} times the

Table 1 Converter and timing-feedback parameters. The applied dwell bound is a simulation assumption, not a hardware-calibrated limit.

Quantity	Value
Source E	30 V
Capacitances C_1, C_2	40 μ F
Inductance L ; resistance R	10 mH; 10 Ω
Intervals N ; period T	9; 286 μ s
Nominal-design dwell bound	22 μ s
Applied dwell bound $d_{\min}$	3 μ s
State scale S_x	diag(10, 20, 1)
Timing scale t_s	10 μ s
LQR state weight Q_n	I_3
Aggressive input weight R_n	0.001 I_8
Conservative input weight R_n	0.1 I_8

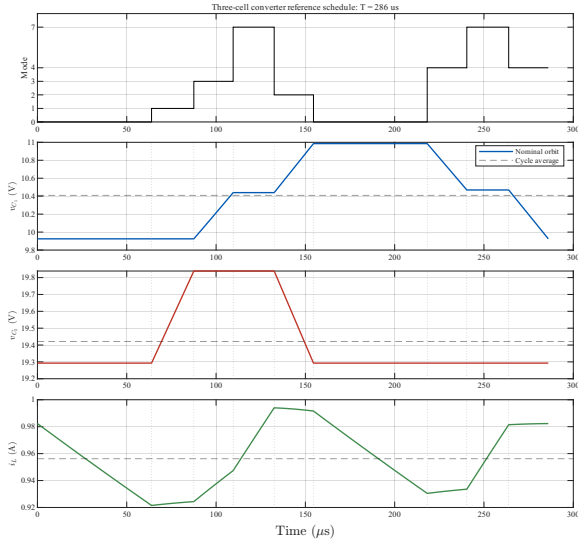


Figure 1 Reconciled reference schedule and periodic state trajectory. Dotted vertical lines mark the eight movable interior switching instants; dashed horizontal lines show cycle averages. The feedback target is the phase-aligned trajectory, not these averages.

diagonal entries of S_x , and the timing step is $10^{-3}t_s = 0.01 \mu\text{s}$. The relative errors for Φ and Γ_τ are 8.518e-12 and 8.344e-11, respectively. For simultaneous normalized state and timing perturbations of amplitude ϵ , we also evaluate

$$r_{\text{lin}}(\epsilon) = \|S_x^{-1}[\mathcal{P}(\mathbf{x}^* + \Delta\mathbf{x}, \delta\tau) - \mathbf{x}^* - \Phi\Delta\mathbf{x} - \Gamma_\tau\delta\tau]\|_2. \quad (33)$$

The fitted log-log slope is 2.000, consistent with the second-order remainder.

For the aggressive gain, the Julia implementation solves (27) using Convex.jl and Clarabel with $\text{tr}(P) = 3$. It returns optimal status and $\eta = 2.911904315 \times 10^{-4}$. The common matrix, displayed to six decimal places, is

$$P \approx \begin{bmatrix} 0.608770 & -0.013731 & 0.027354 \\ -0.013731 & 2.384502 & 0.033889 \\ 0.027354 & 0.033889 & 0.006728 \end{bmatrix}. \quad (34)$$

Full-precision eigenvalue checks give

$$\begin{aligned} \lambda_{\min}(P) &= 4.987946970 \times 10^{-3}, \\ \lambda_{\min}(Q_0) &= 2.911904317 \times 10^{-4}, \\ \lambda_{\min}(Q_1) &= 2.911956096 \times 10^{-4}. \end{aligned} \quad (35)$$

Using the verified $\mu = 2.911904317 \times 10^{-4}$ in (23) gives $r = 0.999938954110$. This worst-case bound accommodates arbitrarily weak correction and is correspondingly slow. It is not a prediction of the aggressive controller's observed settling time.

The Julia solve uses endpoint matrices from a Python reconstruction of the physical model, rather than independently reconstructing the converter. The article pipeline compares those endpoints against its MATLAB model and aggressive gain, then recomputes the inequalities using that model and the returned P . The machine-readable certificate retains full precision; the displayed rounded matrix is checked separately. These are floating-point numerical checks, not interval-arithmetic certification.

5.3 Controlled and open-loop trajectories

All comparison runs start from

$$\mathbf{x}_0 = [7.5143 \quad 20.8211 \quad 0.0314]^T, \quad (36)$$

with normalized error 0.983946. Exact matrix exponentials propagate each actual interval. The simulations do not replace the plant by its linearization. For the continuous-time plots, each interval is sampled at 30 equally spaced local times, including its end, rather than drawing straight lines only between switching states.

Figure 2 compares the physical trajectories over 100 cycles. The controlled trajectory rapidly approaches the reference. Open loop remains visibly displaced over this horizon, despite having

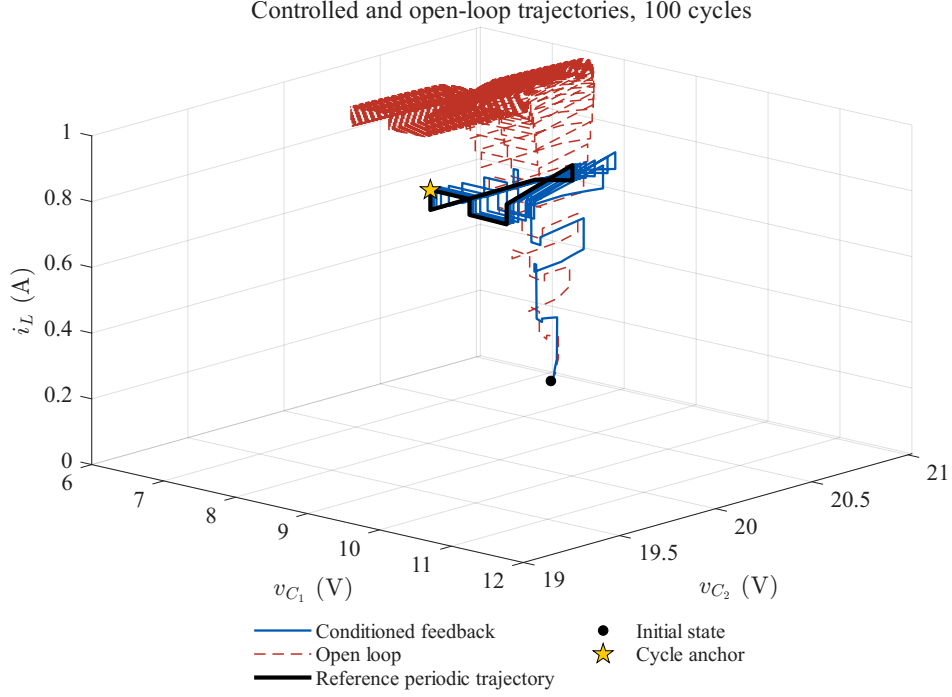


Figure 2 Continuous-time physical-state trajectories over the same 100 cycles, starting from the same initial state and overlaid on one set of axes. Blue shows conditioned aggressive feedback and dashed red shows the fixed nominal open-loop schedule. The black curve is the reference periodic trajectory; the black dot is the initial state and the star marks the cycle anchor. The visible open-loop displacement over this short horizon does not imply instability.

Table 2 Exact simulation results. Threshold cycles for the two feedback laws denote the first sample with $\|S_x^{-1}e_k\|_2 < 0.01$. The open-loop value denotes the crossing after which every remaining simulated sample stays below 0.01.

Quantity	Value
$\rho(\Phi)$, open loop	0.999850
$\rho(A_{cl})$, aggressive	0.568151
Spectral radius, conservative	0.941468
Minimum β ; conditioned cycles	0.293978; 3
Maximum raw offset	65.217 μ s
Maximum applied offset	19.172 μ s
Minimum raw dwell	-43.045 μ s
Minimum applied dwell	3.000 μ s
Minimum conservative dwell	15.502 μ s
Threshold cycle, conditioned	6
Threshold cycle, conservative	20
Persistent crossing, open loop	22009

a Schur cycle matrix. The cycle-start errors in [figure 3](#) quantify the difference without relying on perspective or overlap in the 3D plot.

The aggressive raw law requests a minimum dwell of -43.045 μ s, which would invert an interval. Conditioning acts during 3 cycles, with minimum

factor 0.293978. It reduces the largest absolute offset from 65.217 to 19.172 μ s and keeps every applied dwell at or above 3 μ s. The active constraint reaches this bound, as the maximal-scaling formula predicts.

The conservative law remains feasible without conditioning but crosses the error threshold at cycle 20, compared with cycle 6 for the conditioned aggressive law. [Table 2](#) reports the comparison. These large-error nonlinear trajectories support the controller’s performance in this benchmark; the common- P theorem alone does not certify their entire transient.

5.4 Eventual open-loop convergence

For the same nominal schedule, $\rho(\Phi) = 0.999850 < 1$. Equation (30) therefore predicts eventual open-loop convergence, not merely local convergence of a linear approximation. A separate 100000-cycle exact simulation tests the long time scale hidden by the short comparison.

[Figure 4](#) shows the long-run physical states sampled at the same phase of every cycle. Their damped oscillations approach the components of

Three-cell converter: conditioned large-error response

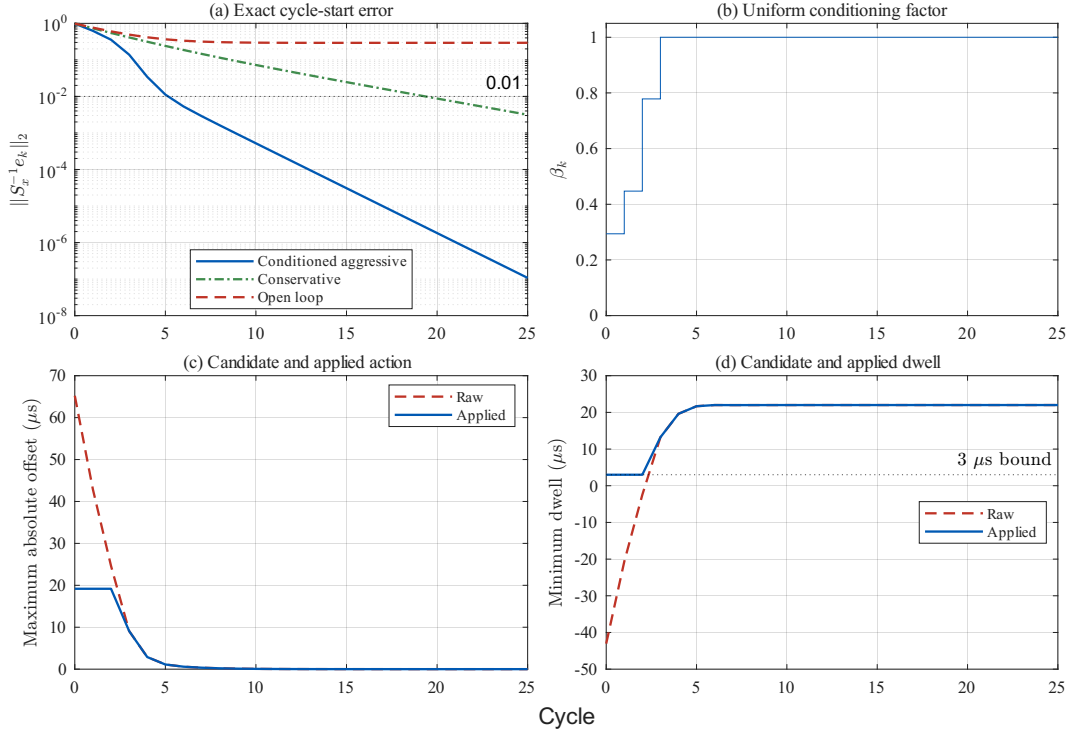


Figure 3 First 25 cycles of the exact comparison. (a) Normalized cycle-start errors for conditioned aggressive LQR, conservative LQR, and open loop. (b) Conditioning factor. (c) Maximum absolute raw and applied switching-instant offsets. (d) Minimum raw and applied dwell durations. Raw actions are candidate commands only; the plant uses the conditioned schedule.

x^* . In the 3D view, selected cycle-start states replace the later within-cycle paths to avoid drawing thousands of overlapping loops. Their approach to the anchor, together with the final cycle overlaid on the reference, shows convergence to periodic motion rather than to a constant physical state.

The normalized cycle-start error remains below 0.01 from cycle 22009 through the end of the run. Since $T = 286 \mu s$, the open-loop transient spans seconds, whereas the controlled threshold crossing occurs within a few milliseconds. The state histories are phase-aligned samples, not continuous-time waveforms or cycle averages.

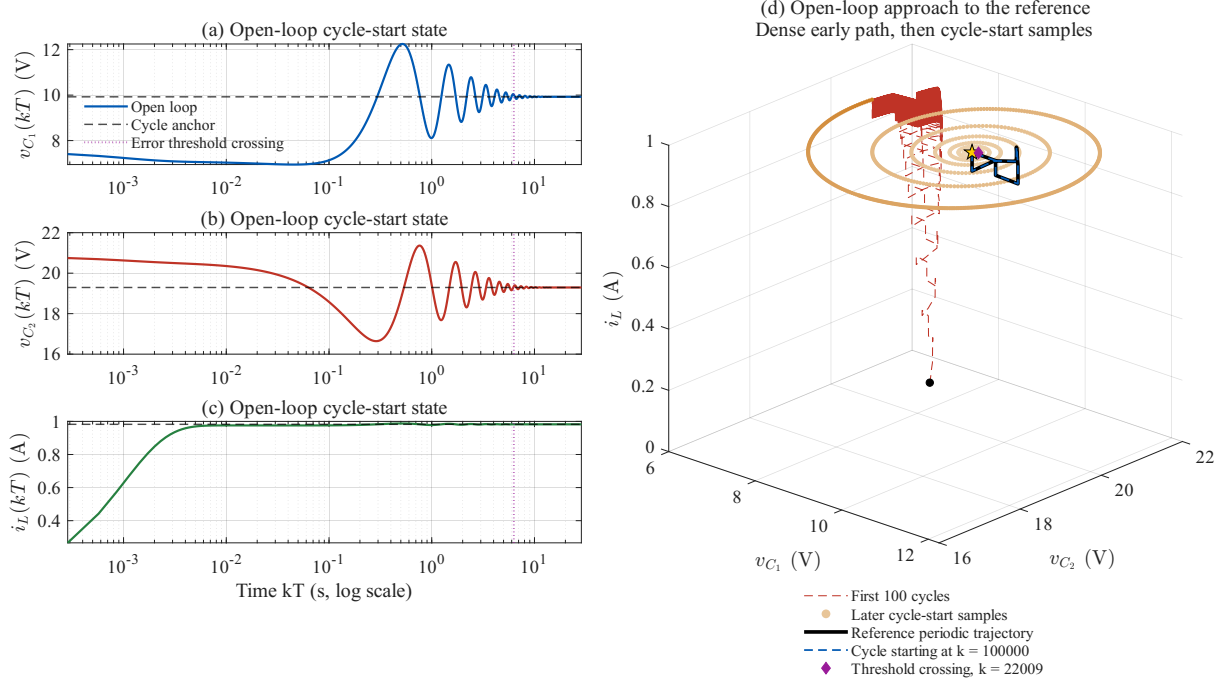


Figure 4 Open-loop convergence over 100000 cycles. (a)–(c) Physical states sampled at cycle starts, plotted against kT on a logarithmic axis; $k = 0$ is omitted. Dashed horizontal lines mark the corresponding components of the reference anchor. The vertical lines mark the persistent normalized-error crossing at cycle 22009. (d) Dense continuous-time motion for the first 100 cycles, followed by selected cycle-start samples that fade with increasing cycle number. Later cycles are represented by their actual start states, not by their full within-cycle paths. The reference trajectory is drawn once, together with the final cycle. The star marks the reference anchor, the black dot the initial state, and the diamond the threshold crossing.

6 Discussion

Uniform conditioning connects two properties that must be distinguished. The dwell inequalities give a closed-form maximal feasible scale for any finite raw action. The common quadratic Lyapunov inequalities then give a sufficient convergence certificate for the associated linearized matrix segment. Both the SDP and the supporting invariant-region computation are offline. Online execution remains a static gain multiplication followed by adjacent differences and a minimum scan.

The one-cycle sensitivity follows Patino et al. [Patino et al. \(2010\)](#), and the common-state-matrix reduction recovers Marcolino et al.’s fixed-period timing model [Marcolino et al. \(2021\)](#). Direction-preserving scaling is also established in control allocation [Nakamura-Zimmerer and Miller \(2026\)](#). The contribution here is the closed-form specialization to coupled dwell inequalities and its

connection to a common- P certificate for conditioned timing feedback, not a new LQR or a new general Lyapunov theorem.

The full-interval certificate is conservative. It must include the slow open-loop endpoint and use the same quadratic function everywhere. It neither proves that every stabilizing timing gain admits such a certificate nor bounds the nonlinear remainder at arbitrary errors. For the benchmark, open loop already approaches the reference, so the benefit of feedback is faster convergence with feasible schedules. Claims of global nonlinear stabilization by the conditioned controller would go beyond these results.

The assumed $3\mu\text{s}$ bound has not been calibrated to hardware. Timer quantization, delays, dead time, implementation margin, state estimation, parameter uncertainty, and experiments remain outside the study. The mode sequence, period, and reference phase are fixed throughout.

7 Conclusion

The proposed conditioner applies the largest dwell-feasible scalar multiple of a proportional timing action in $O(N)$ time, preserving its direction and the fixed cycle period. A common quadratic Lyapunov function found through an offline endpoint SDP certifies uniform exponential stability of the conditioned linearized model for every conditioning-factor sequence in $[0, 1]$. Strict nominal dwell margin also yields a local exact nonlinear stability result. In the converter application, exact 3D trajectories and error histories show that conditioning makes the aggressive action feasible and accelerates convergence relative to an already stable open-loop schedule. The long-horizon simulation confirms convergence to periodic motion. Extending the feedback guarantee beyond a local nonlinear neighborhood requires an explicit remainder or nonlinear Lyapunov analysis.

Appendix A Raw-action invariance

The common- P theorem covers active conditioning in the linearized model. A separate polyhedral calculation answers the narrower question of where the raw action remains feasible forever. Starting from (18), define

$$\begin{aligned} \mathcal{C}_0 &= \mathcal{C}, \\ \mathcal{C}_{j+1} &= \mathcal{C}_j \cap \{\mathbf{e} : A_{\text{cl}} \mathbf{e} \in \mathcal{C}_j\}, \\ \mathcal{C}_\infty &= \bigcap_{j=0}^{\infty} \mathcal{C}_j. \end{aligned} \tag{A1}$$

This is the standard maximal-admissible-set construction [Gilbert and Tan \(1991\)](#). Each finite iterate is polyhedral, and redundant inequalities can be removed offline.

Proposition 2 (Invariant raw-action feasibility). *The set $\mathcal{C}_\infty$ is the maximal positively invariant subset of $\mathcal{C}$ under A_{cl} . If A_{cl} is Schur, every linearized trajectory starting in this set remains raw-action feasible, has $\beta_k = 1$ for all k , and converges to the cycle anchor.*

Proof Membership in $\mathcal{C}_\infty$ is equivalent to $A_{\text{cl}}^j \mathbf{e}_0 \in \mathcal{C}$ for every $j \geq 0$. The conditioner therefore remains inactive,

and no larger subset of $\mathcal{C}$ can have this property. Schur stability gives $A_{\text{cl}}^k \mathbf{e}_0 \rightarrow \mathbf{0}$. $\square$

For the aggressive converter gain, one preimage intersection establishes $\mathcal{C}_\infty = \mathcal{C}$. The computed set has 7 facets, 10 vertices, and physical-error volume $4.5410 \text{ V}^2 \text{ A}$. Every vertex satisfies the raw dwell inequalities within numerical tolerance. Every vertex image has at least $4.347 \mu\text{s}$ dwell margin. Convexity extends these checks to the polyhedron.

The exact nonlinear simulated trajectory begins outside this set and first enters it at cycle 3. This observed entry is not a proof that the nonlinear map reaches or preserves the complete polyhedron. Region vertices and the auxiliary region visualization remain available with the numerical evidence; the common- P theorem does not depend on this calculation.

Declarations

Funding. TODO: name the funding sources and grant numbers, or state that no funding was received.

Competing interests. TODO: state whether the authors have relevant financial or non-financial competing interests.

Data availability. TODO: state where the generated simulation data are available, or that all data supporting the findings are contained in the article. Required for original research.

Code availability. TODO: state where the simulation and figure-generation scripts are available, or that this item is not applicable.

Author contributions. TODO: describe the contribution of each author.

Acknowledgements. TODO: add if applicable, otherwise delete this heading.

ORCID. TODO: add the author ORCID identifiers if available.

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